

AN ORDER-SIX COUNTEREXAMPLE TO THE BRUALDI-LIU EVEN-PERMANENT CONJECTURE

ABSTRACT. Brualdi and Liu conjectured that, for $n \geq 3$, every matrix X in the convex hull Ω_n^{ev} of the even permutation matrices satisfies $\text{per}^{\text{ev}}(X) \geq n!/(2n^n)$. Wanless gave counterexamples for $n = 4$ and $n = 5$ and left open the possibility that the inequality might hold at larger orders. We settle the next order by giving an explicit $M \in \Omega_6^{\text{ev}}$ with $\text{per}^{\text{ev}}(M) = 661/100000 < 5/648$. The inequality also fails on the relative interior of Ω_6^{ev} .

For $\sigma \in \mathfrak{S}_n$, let P_σ be the permutation matrix with a 1 in position $(i, \sigma(i))$, and set

$$\text{per}^{\text{ev}}(X) = \sum_{\sigma \in \mathfrak{A}_n} \prod_{i=1}^n x_{i, \sigma(i)}, \quad \Omega_n^{\text{ev}} = \text{conv}\{P_\sigma : \sigma \in \mathfrak{A}_n\}.$$

Brualdi and Liu conjectured that, for $n \geq 3$ and $X \in \Omega_n^{\text{ev}}$,

$$(1) \quad \text{per}^{\text{ev}}(X) \geq \frac{n!}{2n^n},$$

with equality only at the matrix whose entries are all $1/n$ [1, Conjecture 2]. Wanless disproved (1) for $n = 4$ and $n = 5$ and left open whether it might hold for larger n [2]. We settle the next order.

Theorem 1. *The Brualdi–Liu inequality fails for $n = 6$. More precisely, there exists $M \in \Omega_6^{\text{ev}}$ such that*

$$\text{per}^{\text{ev}}(M) = \frac{661}{100000} < \frac{5}{648} = \frac{6!}{2 \cdot 6^6}.$$

Moreover, (1) fails at points in the relative interior of Ω_6^{ev} .

Proof. Let

$$A = \begin{pmatrix} 3 & 2 & 5 & 0 & 0 & 0 \\ 0 & 3 & 2 & 3 & 0 & 2 \\ 0 & 0 & 3 & 4 & 3 & 0 \\ 2 & 3 & 0 & 3 & 2 & 0 \\ 5 & 0 & 0 & 0 & 0 & 5 \\ 0 & 2 & 0 & 0 & 5 & 3 \end{pmatrix}, \quad M = \frac{1}{10}A.$$

Define

$$\begin{aligned} \pi_1 &= (24)(56), & \pi_2 &= (1234)(56), \\ \pi_3 &= (135), & \pi_4 &= (1345)(26). \end{aligned}$$

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These permutations are even, and direct inspection gives

$$A = 3P_{\pi_1} + 2P_{\pi_2} + 3P_{\pi_3} + 2P_{\pi_4}.$$

Hence $M \in \Omega_6^{\text{ev}}$.

Writing $\text{inv}(\sigma)$ for the inversion number, a direct expansion over the support of A gives exactly six nonzero even terms:

$(\sigma(1), \dots, \sigma(6))$	$\text{inv}(\sigma)$	$\prod_{i=1}^6 a_{i, \sigma(i)}$
$(1, 3, 4, 5, 6, 2)$	4	480
$(1, 4, 3, 2, 6, 5)$	4	2025
$(2, 3, 4, 1, 6, 5)$	4	800
$(2, 3, 4, 5, 1, 6)$	4	480
$(3, 2, 5, 4, 1, 6)$	6	2025
$(3, 6, 4, 5, 1, 2)$	10	800

Thus

$$\text{per}^{\text{ev}}(M) = \frac{480 + 2025 + 800 + 480 + 2025 + 800}{10^6} = \frac{661}{100000}.$$

The strict inequality follows from

$$661 \cdot 648 = 428328 < 500000 = 5 \cdot 100000.$$

Let J_6 be the all-ones matrix. Since

$$\frac{J_6}{6} = \frac{1}{360} \sum_{\sigma \in \mathfrak{A}_6} P_\sigma,$$

for every $t \in (0, 1)$ the matrix

$$M_t = (1 - t)M + t \frac{J_6}{6}$$

has a positive coefficient on every vertex of Ω_6^{ev} , and hence belongs to its relative interior. Since per^{ev} is continuous and $\text{per}^{\text{ev}}(M) < 5/648$, the same strict inequality holds for all sufficiently small $t > 0$. $\square$

REFERENCES

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